Internet Noise: a tale of two subnets

Tim Obezuk

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About Me

- Tim Obezuk
- Solutions Engineer at Cloudflare
- Helping organisations build faster and more secure internet properties

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You already use Cloudflare without knowing it.

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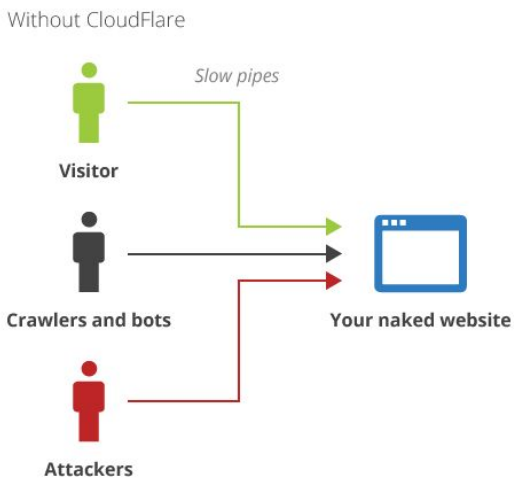
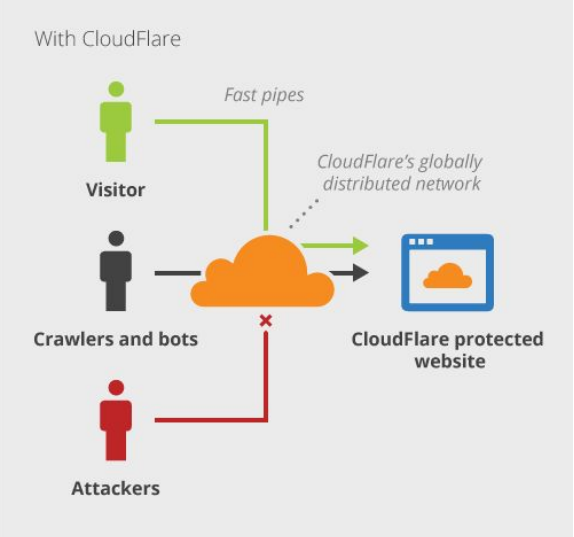
About Cloudflare

Cloudflare makes over 12 million websites faster and safer using our globally distributed Anycast network to deliver essential services to any website

- Performance
- Content
- Optimisation
- Security
- 3rd party services
- Analytics
- Edge Compute





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Some numbers...

- 165+ PoPs and growing across 76+ countries, 25Tbps Capacity
- 195+ Internet exchanges
- > 600bn web requests a day (~10% of all web requests)
- Regular DDoS attacks larger than 500Gbps, 300M PPS
- Largest attack seen 942Gbps
- >130bn DNS requests a day

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Routing History

Routing History

Two interesting IP ranges:

1.1.1.0/24 & 1.0.0.0/24

Bogon prefixes prior to 2010

Known attractor of unwanted traffic such as misconfigurations (proxies, internal use, testing).

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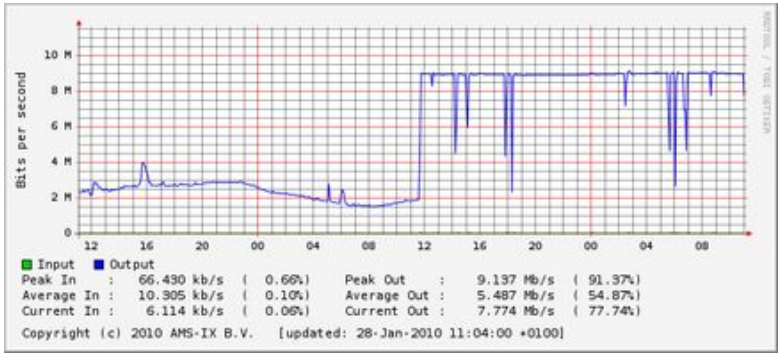
Routing History

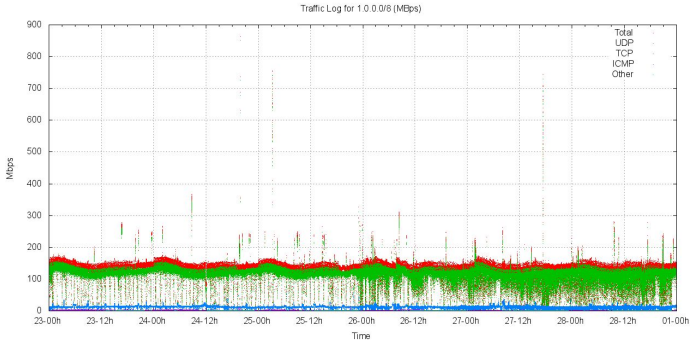
RIPE

- Announced 1.0.0.0/8
- 10Mbps link saturated



Merit

- Announced 1.0.0.0/8
- ~160Mbps - 23 Feb '10 to Mar '10 and collected 7.9TB of pcap



http://www.potaroo.net/studies/1slash8/1slash8.html
https://labs.ripe.net/Members/franz/content-pollution-18

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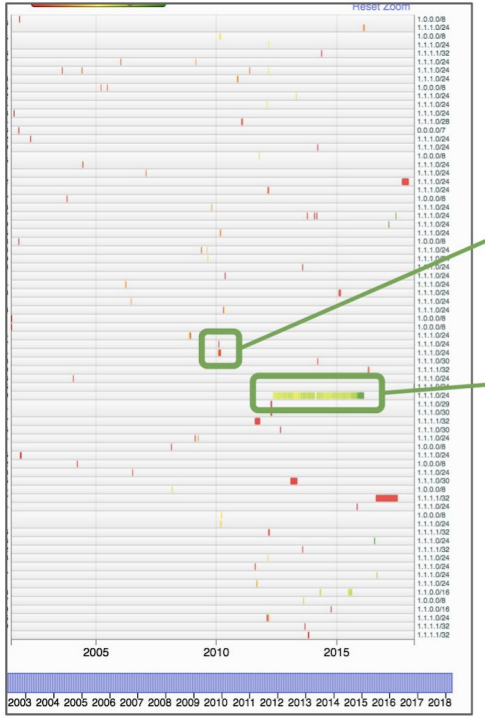
Routing History

YouTube / Google



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APNIC

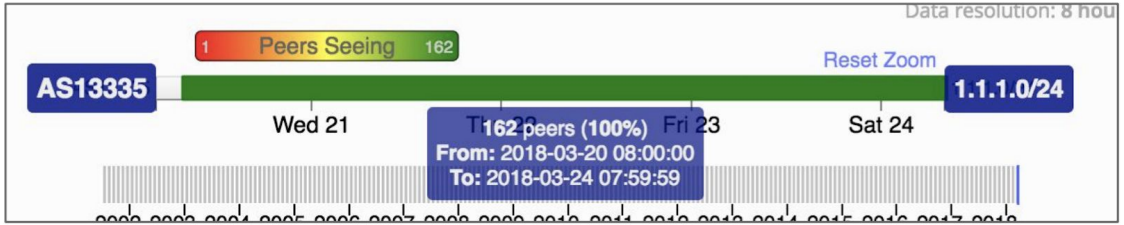
Get IP Manage IP Training Events Research Community

APNIC Labs enters into a research agreement with Cloudflare

By Geoff Huston on 2 Apr 2018

Category: Tech matters

APNIC Labs is partnering with Cloudflare for a joint research project relating to the operation of the DNS.



Routing History

1.1.1.1

Location: World

Type: Raw Performance | Uptime | Quality

Period: Last 30 days

	DNS name	Query Speed
1	1.1.1.1	16.18 ms
2	OpenDNS/Umbrella	22.4 ms
3	Google	34.23 ms
4	SafeDNS	41.31 ms
5	Norton	44.24 ms
6	Neustar	44.8 ms
7	Quad9	46.38 ms
8	Verisign	62.11 ms
9	Comodo	68.28 ms
10	Yandex	126.56 ms

Less is better

It's FAST

It's FREE

Privacy-first DNS Resolver for everyone

APNIC has the IP Address

Cloudflare has the network.

14.84ms in Oceania (dnsperf)

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What’s the Noise / Junk?

Traffic levels

Previous Studies:

- 2010: >100MB/s on 1.0.0.0/8
- 2014: 100Mbps-1Gbps on 1.0.0.0/8

2018:

- 8-13Gb/s
- 1Gb/s solely on 1.1.1.1!

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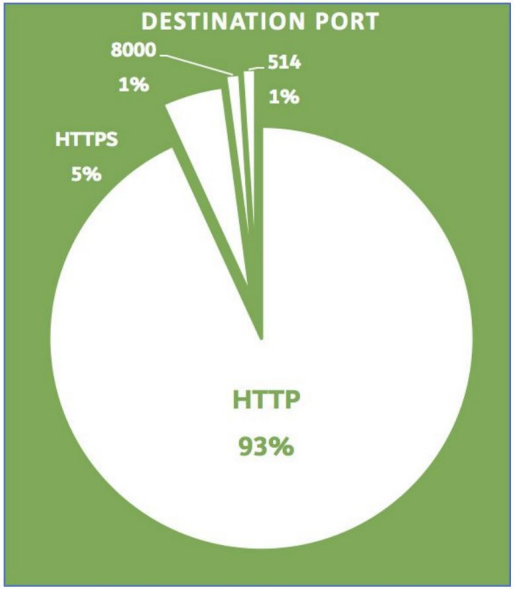
Traffic levels

- TCP traffic (mostly HTTP proxy services)
  - Ports 80, 443, 8000, 8080, 8090, 8765
- UDP traffic (some DNS, syslog)
  - Ports 53, 514, 8000, 80, 8090
- TP-Link DNS 1.0.0.19:
  - https://serverfault.com/questions/365613/tp-link-routers-send-dns-queries-to-1-0-0-19-what-is-that/365630



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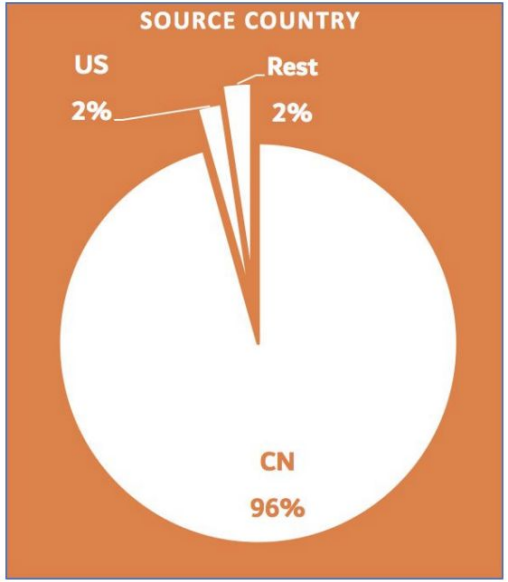
Traffic source

- Aligned with internet populations:
  - Heavily weighted to source from China
  - USA and other large Internet populations



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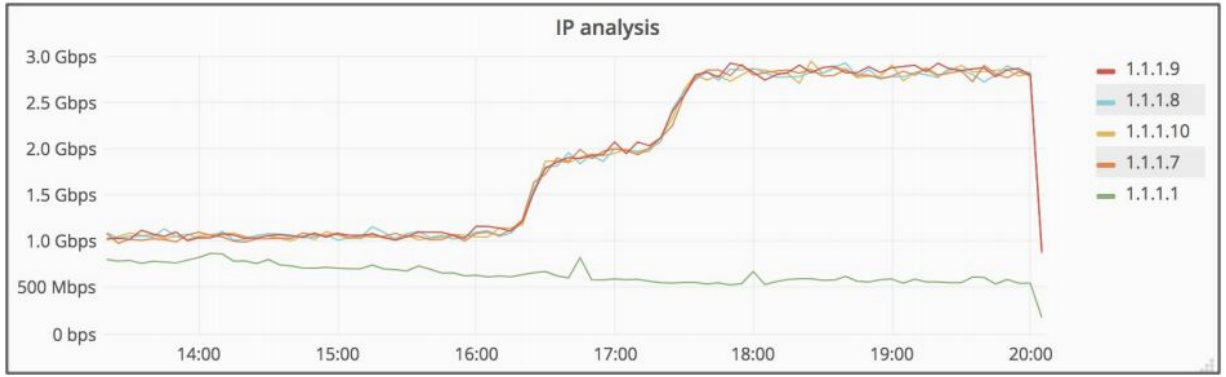
Bursts and patterns

Two increases

- 5 Gb/s → 8 Gb/s between 1600-1715 UTC
- 8 Gb/s → 12.5 Gb/s between 1715-2300 UTC



Mostly on 1.1.1.7, 1.1.1.8, 1.1.1.9 and 1.1.1.10

Destination 80

Increase from China

No particular difference on source IP/net

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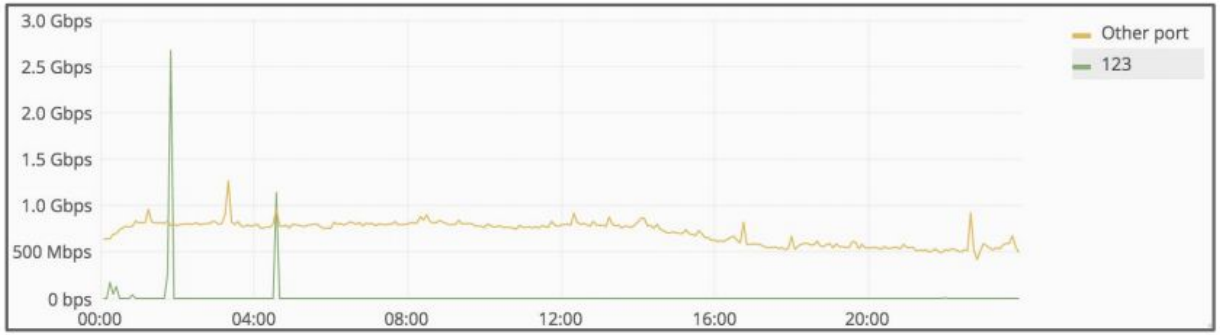
Bursts and patterns

Short bursts:

- Only on 1.1.1.1 between 0100 and 0200 UTC for a few minutes



1-10 gigabits/sec

UDP traffic source 123 (NTP) and 11211 (memcached)

Misconfigured network devices?

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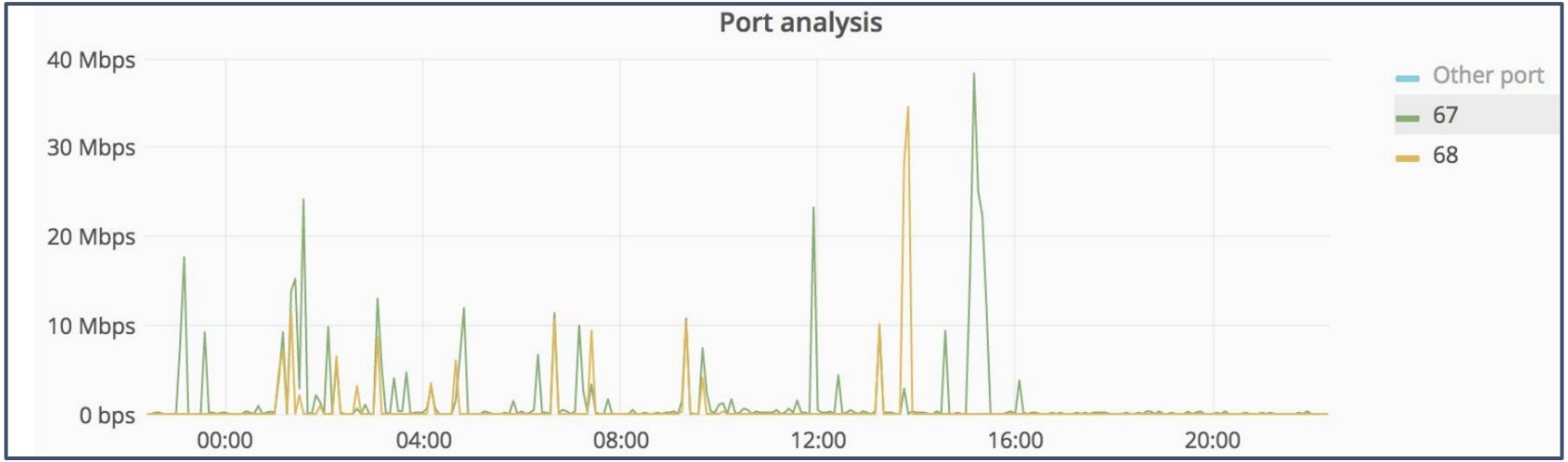
Bursts and patterns

Also **DHCP** spikes. From **Macau**.



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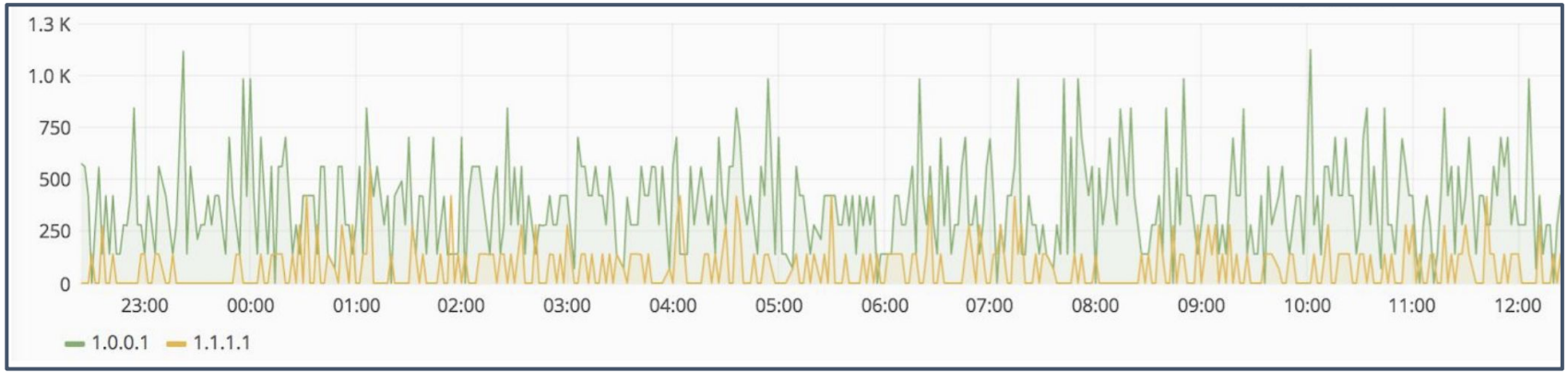
Legitimate Traffic?

Filtering to only UDP/TCP 53, receiving a substantial amount of DNS traffic even before launch.



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What’s changed?

Lots of previous studies into traffic profiles:

- Presentation from 10 years ago at NANOG49
  https://www.nanog.org/meetings/nanog49/presentations/Monday/karir-1slash8.pdf - Merit, APNIC & UMich
- We still see iperf traffic (port 5000/5001)
- Around 10-20 times more traffic than previous studies.

We estimate legitimate traffic to be around 7-13%

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Availability

Availability

Thanks to the Atlas probes, we've run thousands of tests:

Time (UTC)	RTT		Hops	Success	
2018-03-28 11:43	7.504		11	✗	ℹ
2018-03-28 11:43	6.292		11	✗	ℹ
2018-03-28 11:43	6.260		11	✗	ℹ
2018-03-28 11:43	8.558		11	✗	ℹ
2018-03-28 11:43	7.308		11	✗	ℹ
2018-03-28 11:43	3.412		11	✗	ℹ
2018-03-28 11:43	33.123		11	✗	ℹ
2018-03-28 11:43	1.879		1	✓	ℹ
2018-03-28 11:43	21.928		7	✓	ℹ
2018-03-28 11:43	11.641		8	✗	ℹ
2018-03-28 11:43	26.318		4	✓	ℹ

Null-routes

CPE installed in ISP

...

Suddenly an open FTP Server

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Availability

More than **30** major Internet Service Providers all around the world having issues.

- Many null-routing 1.1.1.1/32
- 1.1.1.1/30 is a favorite point-to-point address
- But also using 1.0.0.0/24 for internal purposes (finding devices)
- Most of the ISPs are cleaning their configurations (more than a dozen fixed in less than a week).
- Few non-responses.

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Documentation

Documentation

RFC-5737

```
Internet Engineering Task Force (IETF)                        J. Arkko
Request for Comments: 5737                                    Ericsson
Updates: 1166                                               M. Cotton
Category: Informational                                     L. Vegoda
ISSN: 2070-1721                                                 ICANN
                                                          January 2010

            IPv4 Address Blocks Reserved for Documentation

Abstract

   Three IPv4 unicast address blocks are reserved for use in examples in
   specifications and other documents.  This document describes the use
   of these blocks.
```

192.0.2.0/24
198.51.100.0/24
203.0.113.0/24

Exist for the soul purpose of documentation, diagrams, etc.

HOWEVER...

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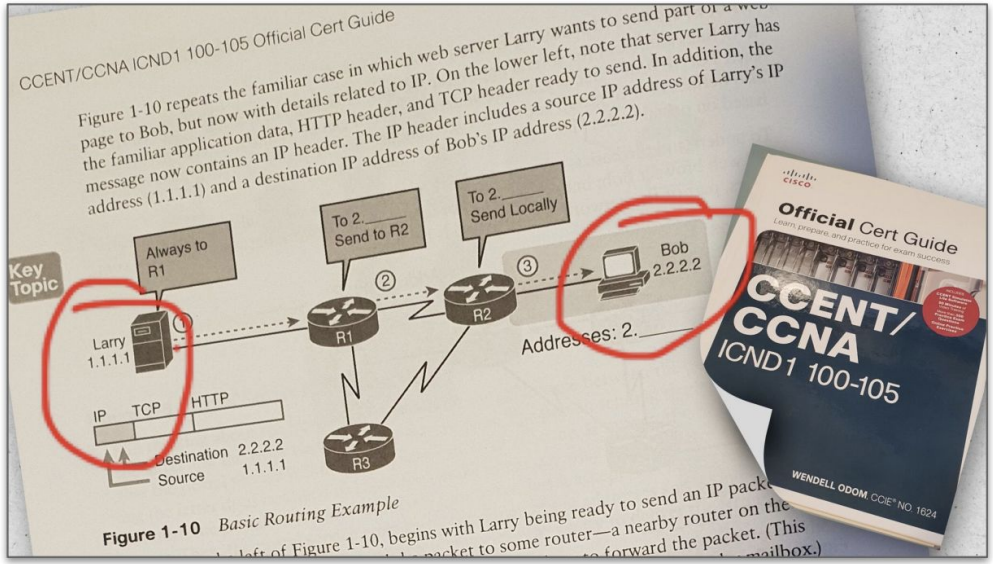
*In addition to performance improvements resulting from the CDN

Documentation

Step 32 In the IP Address text box, enter the IP address of the controller's virtual interface. You should enter a fictitious, unassigned IP address such as 1.1.1.1.



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Just doing it wrong

01-13-2017, 03:44 PM #8

Quote:

Getting tired of typing 192.168. Why doesn't everybody use something simple like 1.1.1.x in a small LAN? What about 0.0.0.x?

I have been using 1.1.1.0/24 subnet for 15+ years on my home LAN and have never found a single instance where any computer in my house ever tried connecting to any address inside the 1.1.1.0-255 range outside my house.

Yes, I realize these are 'publically allocated addresses' but I too got very sick and tired of typing 192.168.blah.blah all the time. I do extensive lab stuff for work where I have servers I build and test in my LAN and am constantly typing IPs all the time.

I still have no regrets about using this subnet. In fact, today in my lab work, I also use 1.1.2.0/24, 1.1.3.0/24, 1.1.4.0/24, 1.1.5.0/24, 1.1.6.0/24, 1.1.7.0/24, 1.1.8.0/24, 1.1.9.0/24 and for the 1.1.2. to 1.1.9. range those are only for lab equipment (have no gateways) for things like iSCSI, vMotion, VSAN and stuff like that so I don't care about them anyway.

You know, if everyone in the world started using 1.1.x.x addresses for home and private LAN use then maybe the industry would change their standard and re-allocate these for official private LAN use, since if someone put a web server on those nobody would ever find their way there. They would be unpopular. Or I guess they are already unpopular because I don't see anyone really using them anyway.

TP-Link routers send DNS queries to 1.0.0.19. What is that?

4

I've got a problem with TP-Link soho routers. The DNS forwarder of those routers tends to ignore the DNS servers obtained by DHCP and instead tries sending all DNS requests to this strange IP: 1.0.0.19? That IP doesn't respond.

Has anyone else seen that happen?

domain-name-system

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Just doing it wrong

Not the first time:

- https://en.wikipedia.org/wiki/NTP_server_misuse_and_abuse

Connectivity testing on TP-Link WiFi extenders [edit]

Firmware for TP-Link WiFi extenders in 2016 and 2017 hardcoded five NTP servers, including Fukuoka University in Japan and the Australia and New Zealand NTP server pools, and would repeatedly issue one NTP request and five DNS requests every five seconds consuming 0,72 GB per month per device.[20] The excessive requests were misused to power an Internet connectivity check that displayed the device's connectivity status in their web administration interface.[20]

- Won't be the last...

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What’s happened in 2019?

bgpstream
@bgpstream

Following

BGP,HJ,hijacked prefix AS13335 1.1.1.0/24, CLOUDFLARENET - Cloudfla,-,By AS263785 Columbus Networks de Cos, bgpstream.com/event/179441

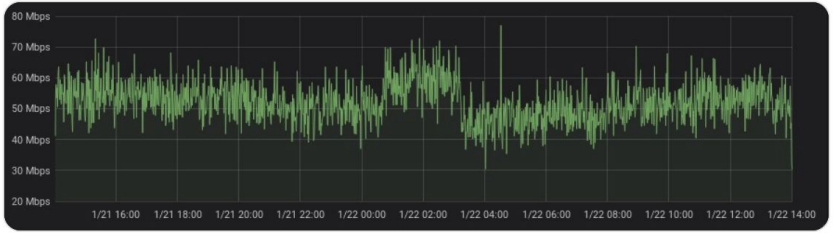
6:00 AM - 16 Jan 2019

majek04
@majek04

Following

Useless fact of the day: The 1.1.1.1 resolver sees about 60Mbps of constant ICMP echo-request traffic.

If you ever left "ping 1.1" in the console, don't worry. You are not the only one.



1:04 AM - 23 Jan 2019

Conclusions

Conclusions

Many different types of misconfiguration

Companies possibly leak their private data:

- Syslog from a government agency
- DHCP data
- Other unknown

We throw away all data, maintain privacy, but not everyone else is nice.

Be vigilant about your own network and follow the best common practices.

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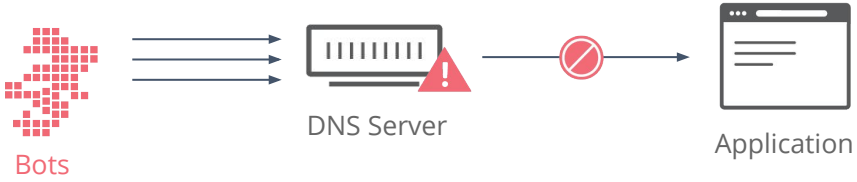
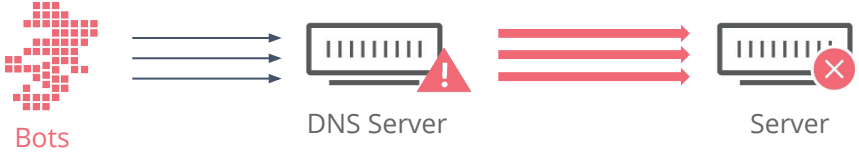
Questions?

Thank you!

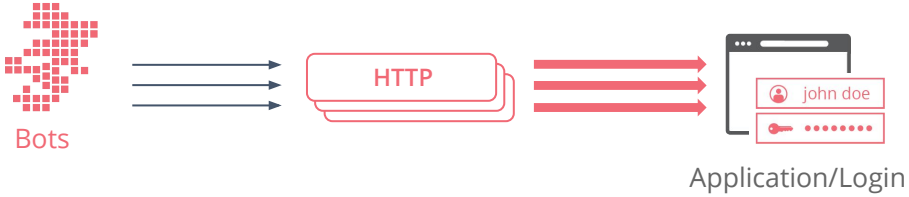
1 Volumetric DNS Flood



2 Amplification (Layer 3 & 4)



3 HTTP Flood (Layer 7)



942Gbps

Largest attack mitigated

500Gbps

300M pps

Common attack size

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